LLM-as-a-Judge Reliability

Phase 3 Combined Presentation

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Category 1: Benchmarks & Frameworks for Judges

Paper 3:
M-Prometheus: A Suite of Open Multilingual LLM Judges

Introduction & Category Alignment

- **Category Context**: Research dedicated to creating standardized testing grounds to evaluate the evaluators themselves.
- **Core Objective**: Quantifying accuracy, consistency, and human-alignment of LLM judges across complex, non-English evaluation tasks.
- **Target Study**: Dissecting M-PROMETHEUS, a suite of open-weight multilingual LLM judges ranging from 3B to 14B parameters.
- **Practical Impact**: Bridging the severe automatic evaluation quality gap between English and non-English target languages.

The Multilingual Evaluation Gap

- **The English-Centric Bias**: Most high-performance LLM-as-a-judge frameworks are optimized exclusively for English text processing.
- **Systematic Disparity**: Lack of reliable non-English evaluators creates a bottleneck, directly hindering the development of robust multilingual models.
- **Baseline Flaws**: Prior open-source multi-lingual attempts fail to support complex pairwise comparisons or inherit brittle capabilities solely from base pretraining.
- **Automatic Metric Failure**: Traditional scalar metrics like BLEU or COMET fail to provide explanatory, text-based reasoning for qualitative decisions.

M-PROMETHEUS Architecture & Training Recipe

- **Model Foundation**: Built by fine-tuning state-of-the-art Qwen2.5-Instruct models across 3B, 7B, and 14B parameter scales.
- **Synthetic Paradigm**: Completely bypasses translated text training data, instead leveraging native multilingual feedback synthesized from scratch.
- **Data Infusion Strategy**: Combines massive custom-generated direct assessment and pairwise preference collections with cross-lingual machine translation eval data.
- **Dual Capability Mode**: Explicitly trained to operate flexibly across both reference-based and reference-free evaluation scenarios.

Methodology Pipeline & Comparison

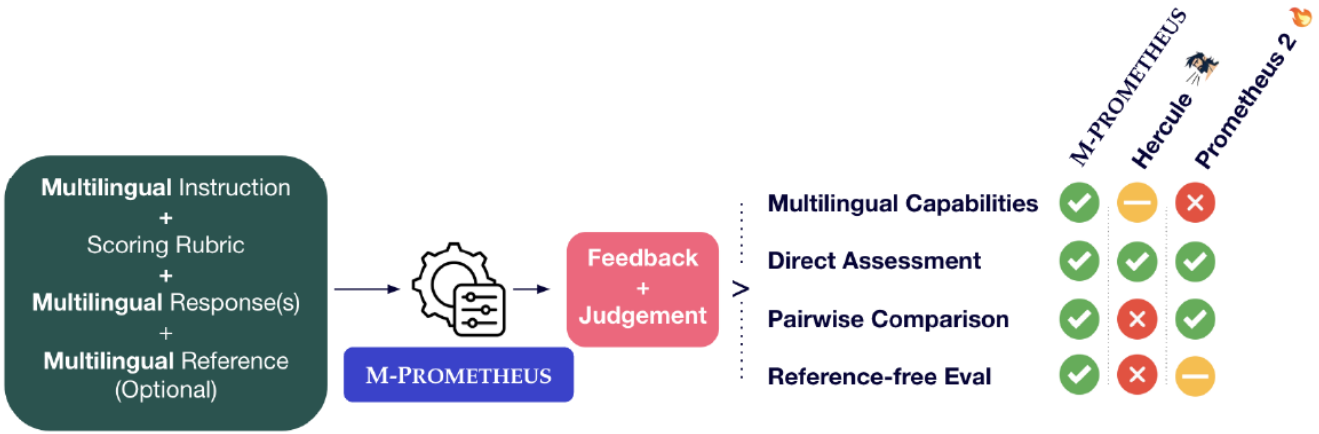


Figure 1: M-PROMETHEUS is a suite of open-weight multilingual LLM judges capable of providing reference-based and reference-free direct assessment and pairwise feedback.

State-of-the-Art Evaluation Results

- **General Benchmark Dominance**: Outperforms all existing open-weight judges on comprehensive multilingual reward benchmarks covering over 20 languages.
- **Surpassing Proprietary Frontiers**: The 14B parameter model successfully outperforms massive closed-source baselines like GPT-4o on MM-Eval.
- **Literary Translation Excellence**: Demonstrates extreme precision on book excerpt translations, a cross-lingual domain where specialized metrics fail.
- **Generative Quality Amplification**: Directly enhances inference-time output accuracy through quality-aware decoding and selection mechanisms.

Core Framework Strengths

- **SOTA Multilingual Benchmarking**: Delivers unmatched evaluation performance against both open-weight and proprietary models in massive non-English ecosystems.
- **Cross-Task Knowledge Transfer**: Demonstrates that integrating machine translation data significantly boosts a judge's general ability to detect language hallucinations.
- **Compute Efficiency at Scale**: Smaller variants like the 3B model deliver massive performance gains relative to their size, making localized deployment practical.
- **Native Capability Preservation**: Drastically enhances multilingual capabilities without degrading core performance or reliability on English benchmarks.

Operational Limitations

- **Diminishing Returns on Scale**: Increasing parameter size to 14B yields structural improvements in theory, but lags behind smaller scales during active decoding tasks.
- **Language Saturation Constraints**: Directly optimized on only 6 language streams, relying heavily on the base model's zero-shot generalization for other target domains.
- **Explanatory Feedback Asymmetry**: While highly reliable at processing non-English prompts and inputs, the default long-form textual rationale remains written in English.

Critical Strategic Outlook

- **Data Engineering Shift**: Establishes native synthetic multi-lingual generation as a fundamentally superior alternative to translating preexisting English datasets.
- **Active Model Development**: Transforms the judge from a passive post-hoc benchmarking script into an active, inference-time text-improvement engine.
- **Future Framework Evolution**: Future reliability paradigms must integrate localized non-English explanation chains and native multi-lingual reasoning paths.

Category 2: Bias Quantification & Fairness Analysis

Paper 5:
Any Large Language Model Can Be a Reliable Judge:Debiasing with a Reasoning-based Bias Detector

The Reliability Bottleneck: Systemic Pitfalls

- **Category Context**: Isolating and quantifying systemic prejudices (position, verbosity, bandwagon, and sentiment biases) that undermine automated trust.
- **Four Structural Vulnerabilities Under Analysis**:
  - **Verbosity Bias**: Prioritizing longer, low-quality answers over concise, mathematically accurate ones.
  - **Position Bias**: Inherent preference for choices based on their order slot.
  - **Bandwagon Bias**: Vulnerability to fabricated majority consensus indicators.
  - **Sentiment Bias**: Distorting scores based on enthusiastic stylistic tone.
- **Flaws of Existing Methods**: In-context prompting cannot break deep-seated cognitive patterns, while direct fine-tuning is impossible for closedendpoints.

Dataset Construction & Distilled SFT Pipeline

- **Source Corpus Foundation**: Built upon curated samples from GSM8K (math), Arena (chat instruction pairs), and ScienceQA (multimodal science).
- **Data Curation Pipeline**:
  - Pairwise sets are modified to introduce targeted skews (e.g., stripping reasoning steps for verbosity, swapping choice order for position).
  - Ground-truth bias labels are mathematically computed by comparing biased evaluations against control judgments.
- **Reasoning Distillation Training Loop**:
  - A high-capacity Teacher LRM generates 1.67k expert reasoning traces explaining the flaws behind the judge's mistakes.
  - Hard validation filtering retains only traces matching the ground-truth label.
  - Base reasoning models (1.5B to 14B scales) undergo Supervised Fine-Tuning (SFT) jointly on all four bias types to learn core logical reasoning.

RBD Framework Pipeline Overview

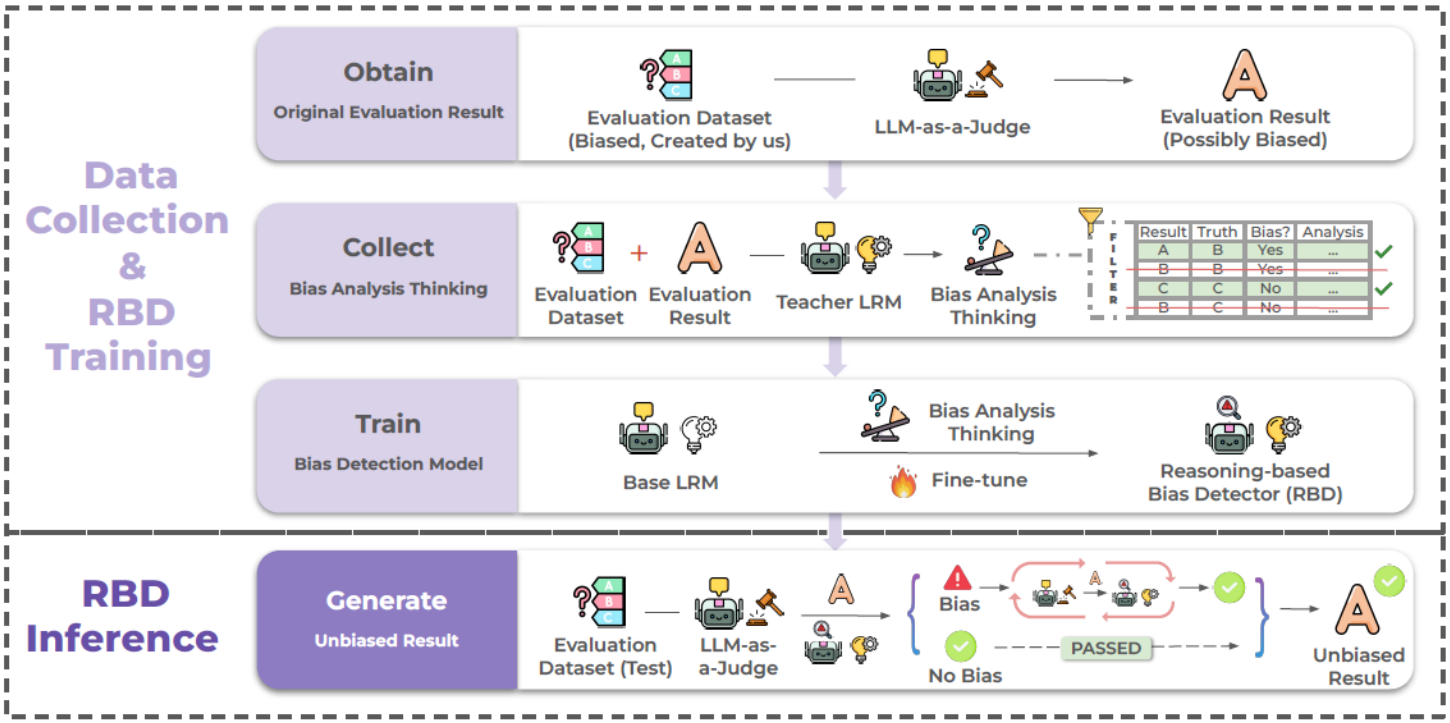


Figure 1: Overview of the Reasoning-based Bias Detector (RBD) framework. During **RBD inference**, it examines biased evaluation results produced by an LLM-as-a-Judge. If bias is identified, RBD generates a reasoning-based bias analysis to guide the LLM in reflecting on and potentially revising its evaluation; otherwise, the original judgment remains unchanged. To train RBD, we design a **data collection and distilled reasoning-based training pipeline**. We first construct a biased dataset containing specific types of bias and collect possibly biased evaluation results from the LLM evaluator. Then, a teacher Language Reasoning Model (LRM) produces bias analysis thinking based on the evaluation context. These analyses are filtered and used to fine-tune a base LRM into the final RBD model capable of identifying and correcting evaluation bias.

Figure 1: The end-to-end framework layout demonstrating dataset generation, validation filtering, and iterative feedback-driven self-correction loops.

Quantitative Evaluation & Key Results

- **Empirical Baseline Improvement**: Evaluated across 8 premier judges (GPT-4o, Claude-3.5, DeepSeek-V3, and LLaMA-3.1 series).
- **Core Performance Metrics**:
  - Integrating RBD-8B boosts evaluation accuracy by an average of **18.5%** and consistency by **10.9%**.
  - Outperforms prompt-based baselines by **12.8%** and out-of-the-box fine-tuned judge models by **17.2%**.
  - Smaller engines see massive jumps; LLaMA-3.1-8B accuracy surges up to **71.8%** on verbosity mitigation.
- **Stress Testing & Scalability**:
  - Scaling the RBD from 1.5B to 14B steadily enhances average accuracy gains (+6.5% to +15.4%).
  - Successfully mitigates multi-bias scenarios, securing recovery gains of +45% and +63.8% under overlapping vulnerabilities.

Core Framework Strengths

- **True Cognitive Understanding**: Avoids exploiting synthetic data shortcuts by enforcing full paragraph-level comparative analysis before outputting labels.
- **Universal Agnostic Integration**: Operates entirely via text, facilitating seamless decoupling and deployment alongside proprietary, black-box commercial APIs.
- **Cross-Domain Generalization**: Maintains peak performance on unseen benchmarks (LLMBar, JudgeBench) and new domains (FactQA) without task-specific training.
- **Negligible Resource Overhead**: Leveraging optimized inference backends achieves an ultra-low latency of 1.5 seconds per sample.

Operational Limitations

- **Cumulative Compute Cost**: While single-sample costs are fractional, multi-round iteration introduces cost aggregation when scaled across millions of requests.
- **Structural Constraint Scope**: Heavily tuned for multiple-choice and pairwise layouts; it does not natively adapt to purely open-ended generation score sheets.
- **Social Blind Spots**: Focuses strictly on surface-level format skews; it cannot explicitly target embedded demographic, cultural, or gender-based social biases.

Critical Strategic Outlook

- **The Scaled Stubbornness Trap**: Larger frontier models possess high internal certainty, making them structurally more stubborn and resistant to external correction.
- **The Authoritative Distillation Flaw**: Distilled, lightweight engines inherit the highly assertive tone of parents, masking an inability to deeply analyze foundational logic.
- **Future Paradigm Re-alignment**: Automated reliability cannot rely on internal alignment; trustworthy grading demands a hard transition toward decoupled, multi-agent referee structures.

Category 3: Scalability Limits & Benchmark Robustness

Paper 4:
Tuning LLM Judge Design Decisions for 1/1000 of the Cost

Title & Overview

- **Paper**: *Tuning LLM Judge Design Decisions for 1/1000 of the Cost*
- **Goal**: Replace expensive human evaluation with open-weight LLM judges — no quality loss, 1/1000 cost.
- **Category**: Scalability Limits & Benchmark Robustness.

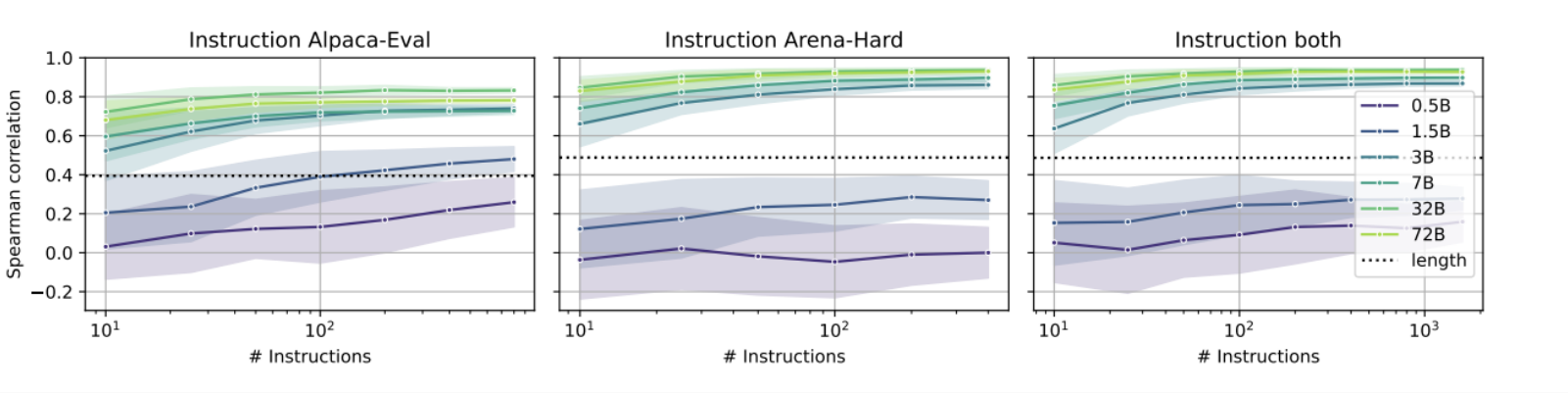
Problem Statement & Current Challenges

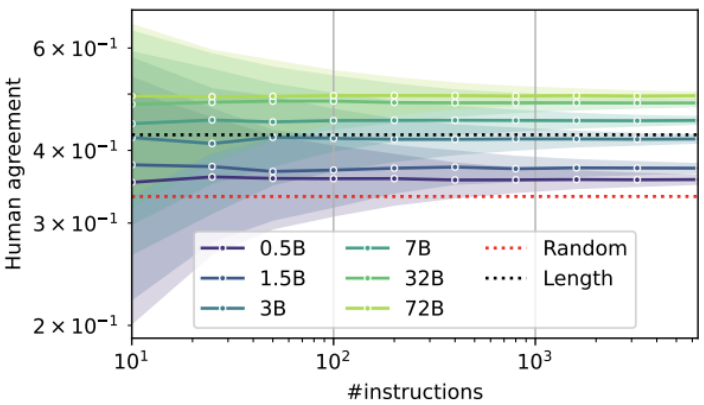
- **GPT-4 dependency**: sanctions risk, sudden behavior shifts, high cost.
- **Confounding factors**: model, prompt, hyperparameters change together (Alpaca-Eval, Arena-Hard).
- **Cognitive biases**: *length*, *position*, *self-enhancement*.

Scaling Laws & Optimal Metric

- **Scaling up alone is not enough**: sub-7B judges act blindly, performing *worse* than the length baseline.
- **Spearman Correlation** (whole-leaderboard vs. human):
  - Noisy and costly; high coefficient of variation (CV) for small models.
  - Grows only with many more instructions — expensive to measure.
- **Human Agreement** (chosen metric):
  - High statistical stability, low noise ($CV \approx 1.3$).
  - Stays *stationary* across instruction counts.
  - Distinguishes large models (32B, 72B) with far fewer, cheaper instructions.
- **Takeaway**: Human Agreement exposes Alpaca-Eval's simplicity vs. Arena-Hard's robustness, at a fraction of the cost.

Scaling Laws — Figure 1 & Figure 2





$\#_{\text{params}}$	Sp. corr. (↑)	CV(↓)	Hum. agr. (↑)	CV(↓)
0.5B	0.09 ± 0.189	207.60	0.36 ± 0.006	1.70
1.5B	0.33 ± 0.137	41.26	0.37 ± 0.006	1.61
3B	0.82 ± 0.066	8.11	0.42 ± 0.006	1.45
7B	0.83 ± 0.082	9.84	0.45 ± 0.006	1.33
32B	0.90 ± 0.052	5.75	0.48 ± 0.006	1.29
72B	0.88 ± 0.074	8.43	0.50 ± 0.006	1.27

Table 1. Comparison of the variability of Spearman-correlation and Human-agreement metrics when using 6500 random annotations and the same default prompt for all model sizes.

Figure 2: Performance scales dynamically with both model size and instruction count,

Hyperparameter Search Space

- **4,480 configurations**.
- **Models (7 open-weight)**: Llama 3.1 (8B, 70B), Qwen 2.5 (7B, 27B, 70B), Gemma 2 (9B, 27B).
- **Temperature**: [0.0, 0.01, 0.1, 1.0] + order-swap averaging.
- **Prompt formats (80)**: single score, Likert, `pair`, multi-criteria; JSON vs. raw text.

Smart Search Methodology

- **Multi-Fidelity (Successive Halving)**:
  - 4,480 $\rightarrow$ 400 instructions.
  - 1,200 $\rightarrow$ 1,200 instructions.
  - 400 $\rightarrow$ 3,548 LMSys instructions.
- **Cost cut to** $\frac{1}{3}$ of the traditional approach.
- **Multi-Objective**: non-dominated sort $\rightarrow$ Pareto front (accuracy vs. token cost).

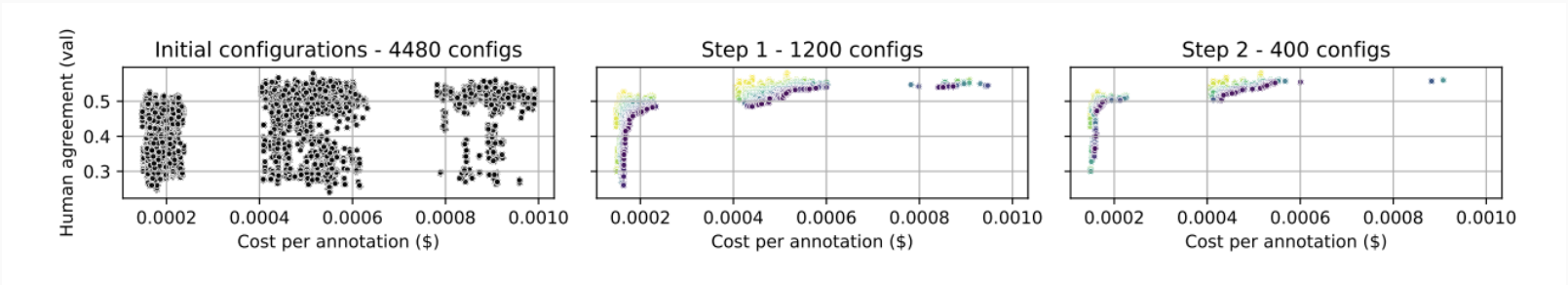


Figure 3: Multi-Fidelity Pareto Optimization (Successive Halving).

Survival Analysis: Winning Configs

- `pair` **format dominates**: 0–10 score per model.
- **CoT hurts**: `explanation/answer` lowers accuracy, raises cost.
- **Temperature** ≈ 0 (0.0, 0.01) wins.
- **Order-swap averaging** boosts stability.

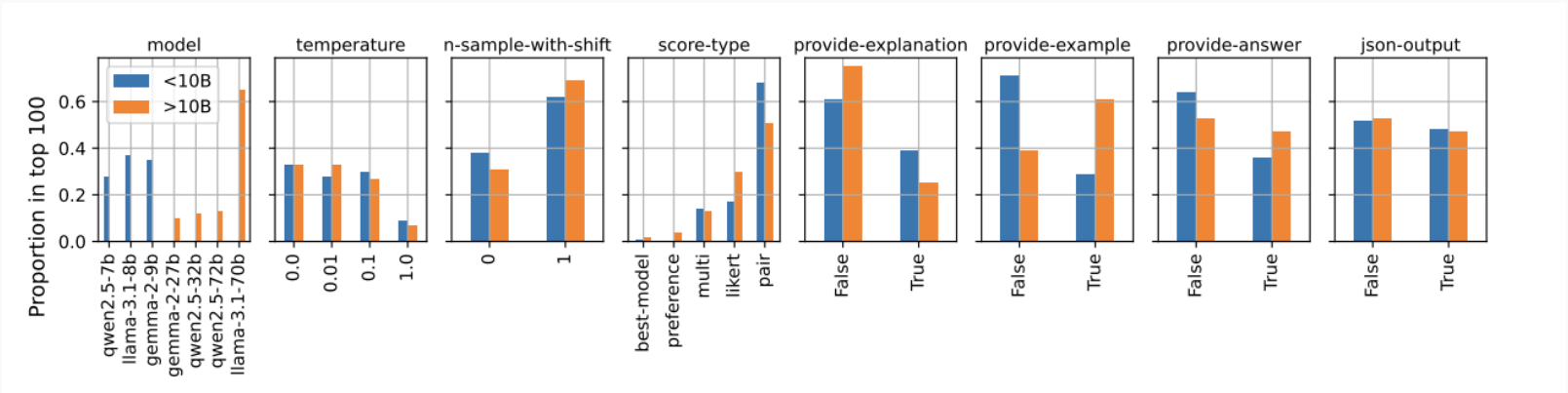


Figure 4: Hyperparameter Survival Analysis (Top 100 Configurations).

Results: LMSys Data

- 3 judges (Small <10B, Medium <32B, Large <72B) on 3,000 test instructions.
- Fine-tuned PandaLM-7B, JudgeLM-7B fail (distribution shift).
- **Ours-large**: Human Agreement 0.49 ≈ GPT-4o-mini (0.50).
- **Cost**: $0.48 vs. $1.2 per 1k.

Judge	Human agr. (↑)	Cost per 1K ann. (↓)
Random	0.33 +/- 0.01	-
Length	0.42 +/- 0.01	-
PandaLM-7B	0.38 +/- 0.01	6.0
JudgeLM-7B	0.42 +/- 0.01	8.6
Arena-Hard	0.50 +/- 0.01	1.2
Ours-small	0.45 +/- 0.01	0.21
Ours-medium	0.47 +/- 0.01	0.48
Ours-large	0.49 +/- 0.01	0.48

Figure 5: Comparison of judges on LMSys test instructions

Results: PandaLM & Arena-Hard

- **PandaLM**: **Ours-small** (<10B) 0.67 > dedicated 70B PandaLM.
- **Arena-Hard**: **Ours-medium** Spearman 0.93 @ $0.48/1k vs. GPT-4 0.90 @ $50/1k.

Judge	Human agr. (↑)
Random	0.33
Length	0.60
GPT-3.5	0.63
GPT-4	0.67
PandaLM-7B	0.62
PandaLM-70B	0.67
Ours-small	0.67
Ours-medium	0.78
Ours-large	0.76

Table 1: Comparison with PandaLM on PandaLM test set

Judge	Sp. corr. (↑)	Cost per 1K ann. (↓)
Length	0.50 +/- 0.21	-
Arena-hard + Claude	0.82 +/- 0.12	75.0
Arena-hard + GPT4	0.90 +/- 0.06	50.0
Ours-small	0.81 +/- 0.10	0.21
Ours-medium	0.93 +/- 0.05	0.48
Ours-large	0.86 +/- 0.09	0.48

Table 2: Spearman correlation between win-rates

Critical Review

- **Limited generalizability**: only 3 Decoder-only families; MoE unproven.
- **CoT rejection uncontrolled**: may be "Lost in the Middle" in sub-10B models, not judging per se.
- **Early-filtering risk**: Successive Halving assumes question homogeneity — hard-task specialists may be wrongly early-stopped.
- **Goodhart's Law**: optimizing for human agreement amplifies hidden human biases.
- **Cost oversimplified**: ignores hardware setup, VRAM, orchestration for 32B/72B.

Paper 6:
Trust or Escalate: LLM Judges with Provable Guarantees

Title & Overview

- **Paper**: *TRUST OR ESCALATE: LLM Judges with Provable Guarantees for Human Agreement*
- **Goal**: Control LLM judging reliability with *mathematically provable* guarantees — not just heuristics.
- **Category**: Scalability Limits & Benchmark Robustness.

Motivation: Why Heuristics Fail

- **Large-scale evaluation** (e.g., 1M samples): human labeling is infeasible.
- **GPT-4 as judge**: expensive heuristic — suffers from systematic bias (length, position) and over-confidence.
- **Key question**: how to make LLM judging *reliable* with a *provable risk bound*?

Mathematical Framework: Trust or Escalate

- **Core idea**: the model should *not* always vote — if uncertain, it should abstain ($\rightarrow$ escalate).
- **Conditional output**:

$$(f_{LM}, c_{LM})(x) = \begin{cases} f_{LM}(x) & \text{if } c_{LM}(x) \geq \lambda \\ \emptyset & \text{otherwise} \end{cases}$$

- f_{LM}: preference vote. c_{LM}: confidence (0–1). λ: threshold.
- **Risk guarantee**:

$$P(f_{LM}(x) = y_{human} \mid c_{LM}(x) \geq \lambda) \geq 1 - \alpha$$

- User sets error risk α and failure probability δ.

Calibration: Fixed-Sequence Testing

- **Calibration set** D_{cal}: small human-labeled set (e.g., 500 samples) to find optimal $\hat{\lambda}$.
- **Binomial upper bound**: model errors on D_{cal} follow a binomial distribution — compute strictest upper bound $\hat{R}^+(\lambda)$ at confidence $1 - \delta$.
- **Fixed-Sequence Testing**:
  - Start from highest threshold ($\lambda = 0.999$) and move down.
  - Stop at last point where $\hat{R}^+(\lambda) \leq \alpha$.
- Much less conservative than Bonferroni correction; maintains higher coverage.

Innovation 1: Simulated Annotators

- **Problem**: standard confidence metrics (predictive probability, verbalized confidence) are over-confident (high ECE).
- **Solution**: simulate diverse human preferences via few-shot prompting.
- **Mechanism**:
  - Query model N times with different in-context examples, each mimicking a specific human j.
  - Ensemble internal vote:

$$c_{LM}(x) = \max_y \frac{1}{N} \sum_{j=1}^{N} p_{LM}(y \mid x; \text{examples}_j)$$

- Disagreement among simulators $\rightarrow$ low confidence $\rightarrow$ escalate. ECE reduced by $\sim$50%.

Innovation 1 : Illustration

Dataset		AlpacaEval				TL;DR			
	Method	Acc.	ECE ↓	AUROC	AUPRC	Acc.	ECE ↓	AUROC	AUPRC
GPT-4-turbo	Predictive Probability	0.724	0.217	0.642	0.852	0.760	0.196	0.731	0.890
	Verbalized Confidence	0.724	0.215	0.550	0.774	0.760	0.194	0.548	0.792
	Randomized Annotators	0.720	0.113	0.705	0.866	0.779	0.079	0.734	0.905
	Simulated Annotators (Maj.)	0.730	0.106	0.718	0.873	0.783	0.062	**0.755**	**0.921**
	Simulated Annotators (Ind.)	**0.734**	**0.095**	**0.723**	**0.877**	**0.788**	**0.039**	**0.755**	**0.921**
GPT-3.5-turbo	Predictive Probability	0.644	0.293	0.581	0.691	0.667	0.228	0.653	0.786
	Verbalized Confidence	0.644	0.306	0.505	0.595	0.667	0.211	0.607	0.716
	Simulated Annotators (Ind.)	**0.694**	**0.058**	**0.632**	**0.793**	**0.725**	**0.043**	**0.704**	**0.842**
Mistral-7B-it	Predictive Probability	0.618	0.374	0.457	0.579	0.661	0.306	0.613	0.735
	Verbalized Confidence	0.618	0.414	0.490	0.627	0.661	0.335	0.578	0.680
	Simulated Annotators (Ind.)	**0.684**	**0.075**	**0.632**	**0.772**	**0.696**	**0.103**	**0.654**	**0.807**

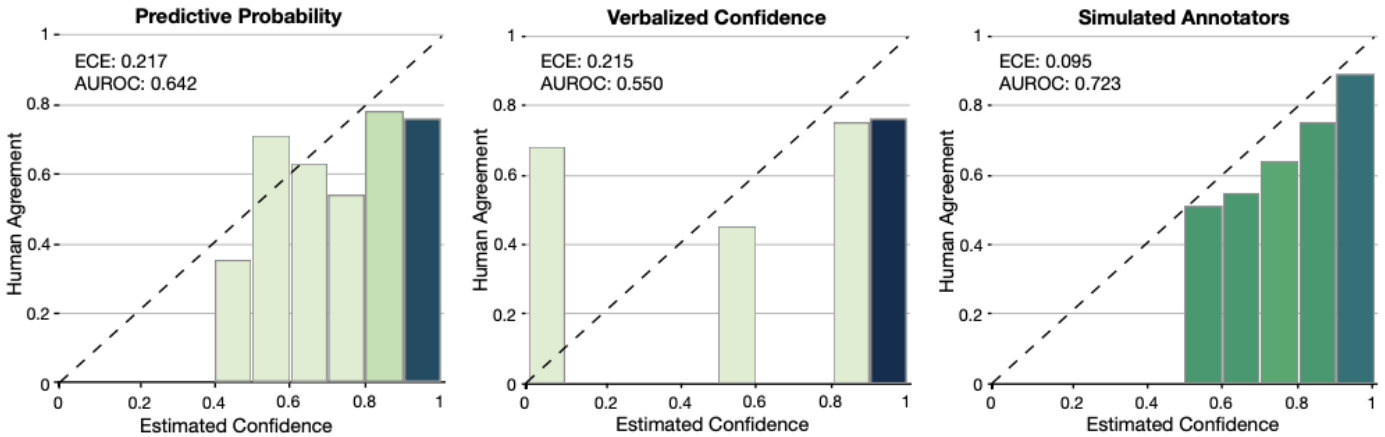


Figure 6: Reliability plot for confidence estimation methods

Innovation 2: Cascaded Architecture

- **Goal**: cut massive cost of multi-testing with large models (GPT-4).
- **Cascade pipeline**:

$$\text{Mistral-7B-it} \longrightarrow \text{GPT-3.5-turbo} \longrightarrow \text{GPT-4-turbo}$$

- Each layer: if $c(x) \geq \lambda_i \rightarrow$ vote and stop; else $\rightarrow$ escalate.
- Abstention policy at each layer auto-tuned based on global risk budget — no manual chain tuning.

Experimental Setup

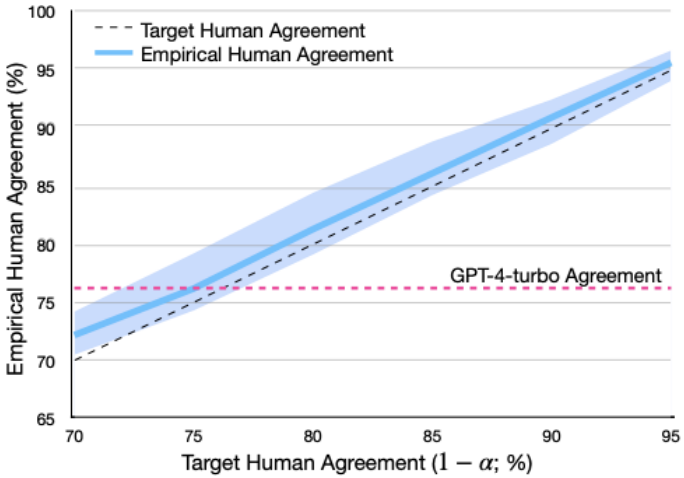
- **Models**: Mistral-7B-it $\rightarrow$ GPT-3.5-turbo $\rightarrow$ GPT-4-turbo.
- **Stats**: $|D_{cal}| = 500, \quad \delta = 0.1$ (90% success guarantee), averaged over 1000 runs.
- **Datasets**:
  - **TL;DR**: summarization task (multiple human labels $\rightarrow$ use independent simulators).
  - **Chatbot Arena**: real-world chat comparisons (single human label $\rightarrow$ majority-based simulators).

Results: TL;DR

- **Risk control**: GPT-4 alone achieves 0% success at 90% agreement target; proposed method passes *all* risk levels.
- **Coverage at 90% agreement**: 55.7% of data covered; half handled by Mistral + GPT-3.5.
- **At 70% agreement**: coverage reaches 96%; Mistral alone handles 84%.

Results: TL;DR

Method	Evaluator Composition (%)			Coverage (%)	Guarantee Success Rate (%)
	Mistral-7B	*GPT-3.5-turbo*	*GPT-4-turbo*		
No Selection	0.0	0.0	100.0	100.0	0.0
Heuristic Selection	0.0	0.0	100.0	89.6	42.0
Cascaded Heuristic Selection	59.6	15.0	25.5	64.6	0.0
Point-Estimate Calibration	100.0	0.0	0.0	5.6	54.7
	0.0	100.0	0.0	9.4	79.0
	0.0	0.0	100.0	57.7	47.5
Cascaded Selective Evaluation	**28.3**	**28.2**	**43.5**	**55.7**	**90.8**



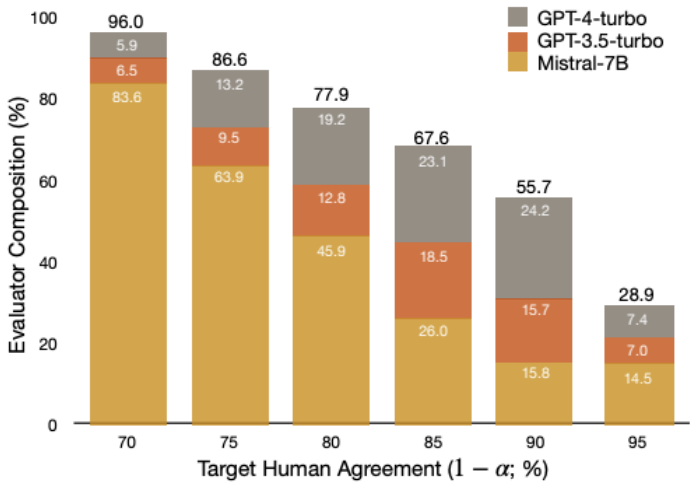


Figure 7: TL;DR results

Results: Chatbot Arena

- **Guaranteed 85% agreement**: system success rate $= 91.0\%$ (matches $\delta = 0.1$ expectation); all baselines near 0%.
- **Total coverage**: 63.2%.
- **Load distribution**: Mistral-7B (23.7%), GPT-3.5 (58.8%), GPT-4 (only 17.5%).
- **Key**: $>$82% of evaluations completed without GPT-4, yet outperform blind GPT-4.

Results: Chatbot Arena

Method	Evaluator Composition (%)			Coverage (%)	Guarantee Success Rate (%)
	Mistral-7B	*GPT-3.5-turbo*	*GPT-4-turbo*		
No Selection	0.0	0.0	100.0	100.0	0.0
Heuristic Selection	0.0	0.0	100.0	95.2	0.1
Cascaded Heuristic Selection	57.1	15.2	27.7	79.7	0.3
Point-Estimate Calibration	100.0	0.0	0.0	0.0	0.0
	0.0	100.0	0.0	40.5	57.2
	0.0	0.0	100.0	60.9	54.4
Cascaded Selective Evaluation	**23.7**	**58.8**	**17.5**	**63.2**	**91.0**

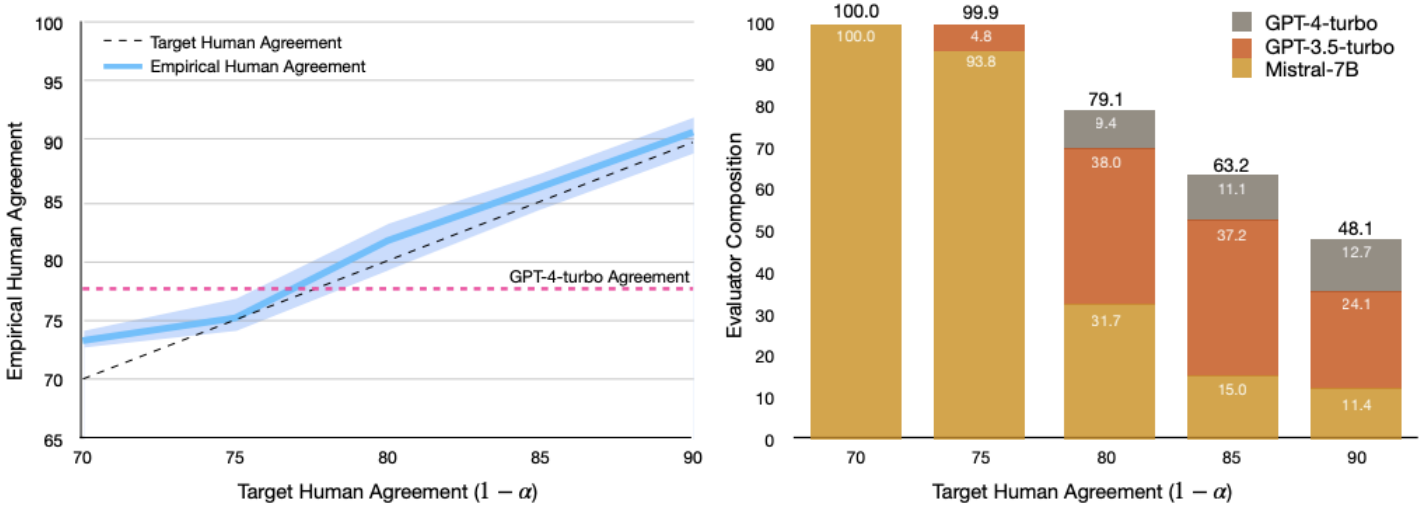


Figure 8: ChatArena results

Critical Review

- D_{cal} **sensitivity**: 500 samples $\rightarrow$ conservative upper bound $\rightarrow$ reduced real coverage; hidden human-labeling cost.
- **Token overhead**: Simulated Annotators with N judges $\times$ K few-shot examples $\rightarrow$ huge context load even for cheap models.
- **Binary pairwise only**: the binomial bound does not extend to scalar scoring (e.g., 1–10 ratings).
- **Mild distribution-shift assumption**: stability under extreme shift (summarization $\rightarrow$ coding/math) unproven; i.i.d. violation collapses the guarantee.
- **Escalation cost ambiguity**: unclear how $\emptyset$ samples are handled; human re-labeling cost can erode financial gains.

Category 4: Structured Evaluation Protocols

Paper 1:
RocketEval: Efficient Automated LLM Evaluation via Grading Checklist

RocketEval: Motivation

- **Problem**: GPT-4-class judges are useful but expensive, API-dependent, private, and hard to reproduce.
- **Observation**: Small open models fail mostly because they miss detailed analysis, not because they cannot answer simple checks.
- **Core Idea**: Convert one broad evaluation into many narrow checklist questions.
- **Goal**: Make a lightweight local judge reliable enough for scalable open-ended evaluation.

Why Small Judges Fail

- Direct CoT judging still leaves weak models uncertain and inconsistent.
- Two dominant error sources are **high uncertainty** and **position bias**.
- RocketEval reduces the task from "give a global score" to "verify checklist items."
- This makes judging more interpretable: every final score has item-level evidence.

RocketEval Framework

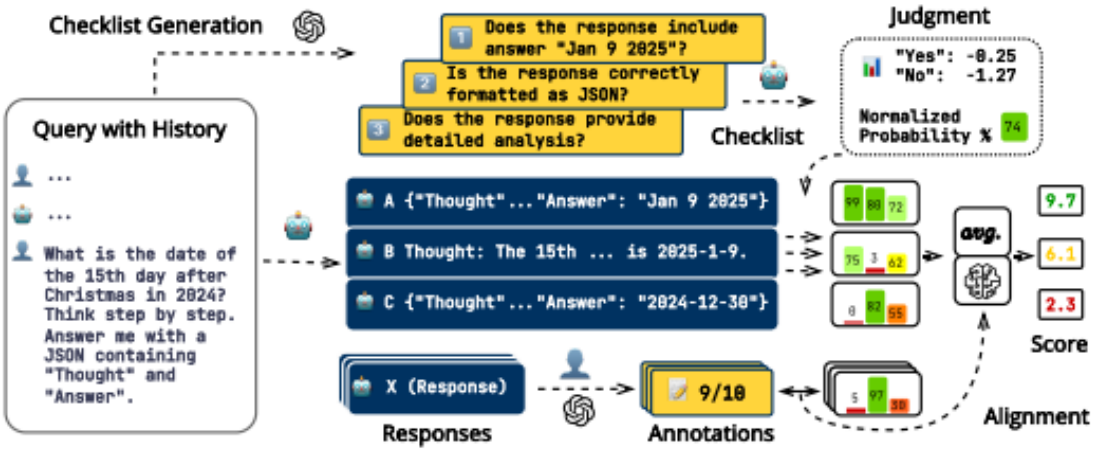


Figure 2: Illustration of **RocketEval** framework for automated LLM evaluation. The framework consists of three components: Checklist Creation, Checklist Grading and Score Prediction.

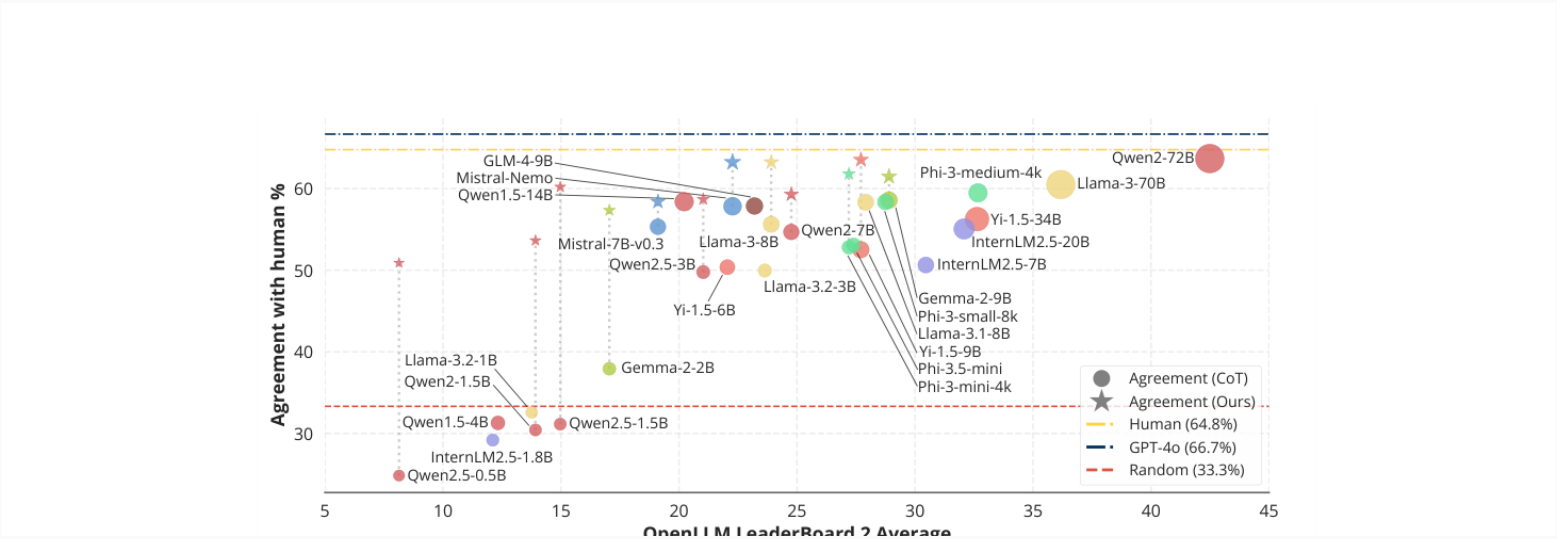
- Three stages: checklist creation, checklist grading, score prediction.
- The pipeline separates *criterion design* from *criterion verification*.

Method: Checklist QA Instead of Global Judgment

- **Checklist creation**: generate focused criteria for correctness, completeness, relevance, and instruction-following.
- **Checklist grading**: ask the lightweight judge whether each criterion is satisfied.
- **Score prediction**: average item scores, or learn weights from supervised labels.

$$\hat{p} = \frac{p_\theta(\text{Yes} \mid x, y, c)}{p_\theta(\text{Yes} \mid x, y, c) + p_\theta(\text{No} \mid x, y, c)}$$

Key Result: Small Judges Become Competitive



- RocketEval improves many small judges over ordinary CoT judging.
- With Gemma-2-2B, it reports **0.965** correlation with human preferences and over **50**× cost reduction.

Critical Review & Bridge to Paper 2

- **Strength**: transparent item-level evidence; cheaper and more reproducible than closed judge APIs.
- **Limitation**: checklist quality still depends on a strong generator; weak or biased checklists can mislead the judge.
- **Limitation**: supervised reweighting still needs reliable labels.
- **Bridge**: Paper 1 improves *structured criteria*; Paper 2 improves *comparative reasoning coverage*.

Paper 2:
Crowd Comparative Reasoning Unlocking Comprehensive Evaluations for LLM-as-a-Judge

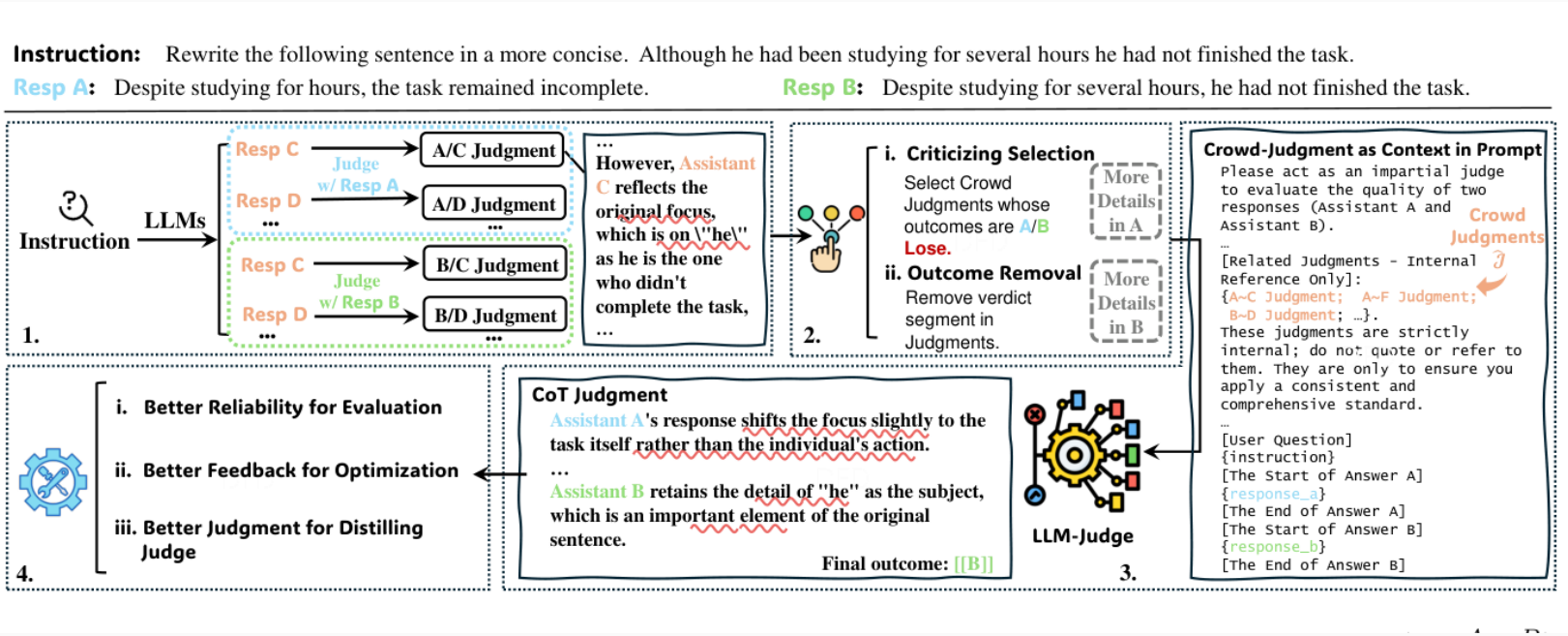
Paper 2: Motivation

- **Problem**: LLM judges often produce incomplete CoT judgments when comparing only two candidate answers.
- **Why it matters**: shallow reasoning misses subtle differences and can produce unreliable preferences.
- **Existing fixes**: majority voting and longer rubrics help, but they are either costly or response-unaware.
- **Main Question**: Can extra crowd responses expose hidden details in the two candidate responses?

Core Idea: Crowd-Based Comparative Evaluation

- Generate several **crowd responses** for the same instruction.
- Compare each candidate response against these crowd responses.
- Use the resulting **crowd judgments** as context for the final A/B judgment.
- The crowd acts as evaluation anchors: different responses reveal different strengths and weaknesses.

CCE Framework



- Crowd judgments are selected, cleaned, and inserted into the final judge prompt.
- The final judge produces a richer CoT and a more reliable preference label.

Main Results Across Judge Benchmarks

Model	Reward Bench	HelpSteer2	MTBench Human	Judge Bench	EvalBias	Avg.
GPT-4o						
Vanilla	85.2	66.1	82.1	66.3	68.5	73.6
LongPrompt	86.9	67.3	81.8	63.5	70.5	74.0
EvalPlan	88.7	65.5	81.4	62.9	74.4	74.6
16-Criteria	87.3	69.1	82.8	66.6	73.7	75.9
Maj@16	87.9	68.9	82.4	68.6	75.5	76.7
Agg@16	88.1	68.7	82.6	67.2	77.9	76.9
CCE-random@16	91.2	69.5	83.1	68.9	80.1	78.6
CCE@16	**91.8**	**70.6**	**83.6**	**70.4**	**85.0**	**80.3**
Qwen 2.5 7B-Instruct						
Vanilla	78.2	60.7	76.1	58.3	57.4	66.1
CCE@16	**80.4**	**64.2**	**76.7**	**64.0**	**79.4**	**72.9**
Qwen 2.5 32B-Instruct						
Vanilla	87.4	**72.3**	79.0	68.9	71.1	75.7
CCE@16	**90.8**	72.1	**82.1**	**70.6**	**80.5**	**79.2**
Qwen 2.5 72B-Instruct						
Vanilla	85.2	**69.5**	79.5	68.3	68.5	74.0
CCE@16	**93.7**	68.5	**88.9**	**75.7**	**85.9**	**82.7**
Llama 3.3 70B-Instruct						
Vanilla	86.4	70.4	81.1	67.1	70.6	75.1
CCE@16	**91.7**	**71.3**	**83.5**	**69.7**	**79.2**	**79.1**

- CCE improves average accuracy across five benchmarks.
- The paper reports an average **6.7%** gain over strong judging baselines.

Beyond Evaluation: Distillation and SFT

- **Judge distillation**: longer, higher-quality CCE rationales train smaller judge models more effectively.
- **Crowd rejection sampling**: CCE helps select better training samples from candidate response pools.
- **Scaling behavior**: accuracy improves as more crowd comparisons are used at inference time.
- **Practical value**: CCE is not only an evaluator; it also improves training data selection.

Critical Review & Relation to Paper 1

- **Strength**: directly targets incomplete CoT reasoning by forcing comparison against diverse anchors.
- **Strength**: useful for evaluation, judge distillation, and rejection sampling.
- **Limitation**: extra crowd responses increase inference cost and prompt length.
- **Limitation**: if crowd responses are low-quality or homogeneous, they may add noise instead of insight.
- **Takeaway**: RocketEval decomposes criteria; CCE expands comparison context. Together, they make LLM judges more structured and more informative.